

Preliminaries

We define the natural numbers $\mathbb{N}$ to be the strictly positive integers. That is, $\mathbb{N} = \{n \in \mathbb{Z} : n > 0\}$. We additionally define $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.

Unless otherwise stated, all rings are commutative and contain a 1.

Given a ring R and two ideals I and J of R , define their sum and product as follows:

$$I + J = \{a + b : a \in I \text{ and } b \in J\},$$

and

$$IJ = \left\{ \sum_{k=1}^n a_k b_k : a_k \in I, b_k \in J, \text{ and } n \in \mathbb{N} \right\}.$$

We can generalize the product of ideals to finite collections of ideals. Let $I_1, I_2, \dots, I_m$ be ideals of a ring R , then we define their product as follows:

$$I_1 I_2 \cdots I_m = \left\{ \sum_{k=1}^n \left(\prod_{l=1}^m a_{k,l} \right) : a_{k,l} \in I_k \text{ and } n \in \mathbb{N} \right\}$$

As a special case, the power ideal I^m is defined for $m \in \mathbb{N}$ as follows:

$$I^m = \left\{ \sum_{k=1}^n \left(\prod_{l=1}^m a_{k,l} \right) : a_{k,l} \in I \text{ and } n \in \mathbb{N} \right\}$$

Given a ring R and an arbitrary indexing set I , we define $R^{(I)}$ as the set of collections $\{a_i\}_{i \in I}$ of elements of R such that $a_i = 0$ for almost all $i \in I$. We can endow this set with an R -module structure by defining componentwise addition and scalar multiplication.

We define $\delta_{i,j}$ as follows:

$$\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Additionally, for the R -module $R^{(I)}$, we define the elements $e_i = \{\delta_{i,j}\}_{j \in I}$ and denote $\{e_i\}_{i \in I}$ as the canonical basis for $R^{(I)}$.

Given an R -module M , it is understood that 0 serves as the identity element of the abelian group M . Additionally, we denote $L(M, A)$ as the set of linear forms on M . That is, $L(M, R)$ is the set of functions $f : M \rightarrow R$ such that $f(m+n) = f(m) + f(n)$ and $f(am) = af(m)$ for all $m, n \in M$ and $a \in R$.

1.1 Divisibility in Principal Ideal Rings

Proposition 1.1: $\mathbb{Z}$ is a principal ideal domain.

Proof: Let I be a nonzero ideal in $\mathbb{Z}$, then we can choose a nonzero element $b \in I$. If $b < 0$, then $b^2 > 0$ and I has a positive element, thus without loss of generality we can assume $b > 0$. By the well ordering principle, we can also assume that b is minimal among the positive integers in I . Now, fix an arbitrary element $x \in I$. Since $\mathbb{Z}$ is a Euclidean domain, we have unique integers q and r such that $x = qb + r$, and where $0 \leq r < b$. Consequently, we have that $r = x - qb \in I$ since I is closed under addition. If $r > 0$, then we have found a positive integer in I that is less than b , a contradiction to our assumption, hence we must have $r = 0$. This shows that all elements in I are multiples of b , and thus $I = (b)$ and is a principal ideal. ■

Proposition 1.2: If K is a field, then $K[X]$ is a principal ideal domain.

Proof: Fix a nonzero ideal I in $K[X]$. Let d be the minimal nonzero degree of any element in I , and let b be an element in I with degree d . Letting x be any element in I , we can use the fact that $K[X]$ is a Euclidean domain to find unique elements $q, r \in K[X]$ where $x = qb + r$, and where $0 \leq \deg(r) < \deg(b)$. By closure, we have that $r = x - qb \in I$, which combined with our assumption on $\deg(b)$ forces $\deg(r) = 0$. Thus, any element $x \in I$ is a multiple of b , meaning that $I = (b)$ and is thus principal. ■

1.2 Diophantine Equations

Theorem 1.3: We have that $x, y, z \in \mathbb{N}$ satisfy the equation $x^2 + y^2 = z^2$ if and only if there exists an integer d and relatively prime integers u and v such that, after possible rearrangement of x and y :

$$x = d(u^2 - v^2), \quad y = 2d uv, \quad \text{and} \quad z = d(u^2 + v^2).$$

Proof: The backwards direction is easy to see, as

$$\begin{aligned} x^2 + y^2 &= [d(u^2 - v^2)]^2 + (2d uv)^2 = d^2(u^4 - 2u^2v^2 + v^4) + d^2(4u^2v^2) \\ &= d^2(u^4 + 2u^2v^2 + v^4) = d^2(u^2 + v^2)^2 = z^2. \end{aligned}$$

Conversely, let $x, y, z \in \mathbb{N}$ satisfy the equation. We can, without loss of generality, assume that they are pairwise relatively prime, since otherwise we can divide out by their gcd. ▲

Theorem 1.4: There exist no integers $x, y, z \in \mathbb{N}$ that satisfy the equation

$$x^4 + y^4 = z^2.$$

Proof: ▲

Corollary 1.5: There exist no integers $x, y, z \in \mathbb{N}$ that satisfy the equation

$$x^4 + y^4 = z^4.$$

Proof: ▲

1.3 Lemmas on Ideals and Euler's ϕ -function

Proposition 1.6: Fix natural numbers q and n , and denote $\tilde{q}$ as the residue class $q + n\mathbb{Z}$. We have that the following are equivalent:

- (a) $\gcd(q, n) = 1$
- (b) $\tilde{q}$ is a unit in the ring $\mathbb{Z}/n\mathbb{Z}$
- (c) $\tilde{q}$ generates the additive group $\mathbb{Z}/n\mathbb{Z}$

As an immediate consequence, we have that $\phi(n)$ is equal to the number of units in the ring $\mathbb{Z}/n\mathbb{Z}$, as well as the number of generators of the additive group $\mathbb{Z}/n\mathbb{Z}$.

Proof: We will prove that (a) $\implies$ (b) $\implies$ (c) $\implies$ (a). We also choose q to be the unique representative of $\tilde{q}$ where $0 \leq q \leq n - 1$.

First, assume (a). By Bezout's lemma, there exist integers x and y such that $qx + ny = 1$. We thus have that $qx \equiv 1 \pmod{n}$, which is equivalent to $\tilde{q} \cdot \tilde{x} = \tilde{1}$. This proves (a) $\implies$ (b), and so we now assume that $\tilde{q}$ is a unit in $\mathbb{Z}/n\mathbb{Z}$. Choose $\tilde{x} \in \mathbb{Z}/n\mathbb{Z}$ such that $\tilde{q} \cdot \tilde{x} = \tilde{1}$, then for arbitrary $\tilde{a} \in \mathbb{Z}/n\mathbb{Z}$ we have $\tilde{a} = \tilde{a} \cdot \tilde{x} \cdot \tilde{q}$. This is equivalent to $\tilde{a} = (ax)\tilde{q}$, which shows that $\tilde{q}$ generates all elements in $\mathbb{Z}/n\mathbb{Z}$, and (b) $\implies$ (c). Finally, assume $\tilde{q}$ to be a generator for $\mathbb{Z}/n\mathbb{Z}$, then there exists an integer x where $x\tilde{q} = \tilde{1}$, but this means that $xq \equiv 1 \pmod{n}$. We can choose an integer y such that $xq - 1 = yn$, which implies that $xq - yn = 1$. By Bezout's lemma, this implies that $\gcd(q, n) = 1$, thus showing (c) $\implies$ (a) and completing the proof. ■

Lemma 1.7: Let R be a ring and let I and J be ideals of R such that $I + J = R$, then we have that $I \cap J = IJ$ and $R/IJ \cong R/I \times R/J$.

Proof: We have that $IJ \subseteq I$ since I absorbs multiplication from J . Similarly, $IJ \subseteq J$, and thus $IJ \subseteq I \cap J$. Fix an element $x \in I \cap J$. By assumption, $I + J = R$, and so

there exist elements a and b in I and J , respectively, such that $a + b = 1$, hence we obtain the equality $x = xa + xb$. This means that x is an element of IJ , and thus $IJ \subseteq I \cap J$. As a result, we have $IJ = I \cap J$.

Next, we consider the ring homomorphism $\theta : R \rightarrow R/I \times R/J$, which we define as $\theta(a) = (a + I, a + J)$. Since $a + I = I$ and $a + J = J$ if and only if $a \in I \cap J$, we have that $\ker \theta = I \cap J = IJ$. This shows that, if $a + IJ$ and $b + IJ$ are equivalent elements of R/IJ , then $\theta(a) = \theta(b)$. We can thus define a function $\phi : R/IJ \rightarrow R/I \times R/J$, defined as $\phi(a + IJ) = \theta(a) = (a + I, a + J)$. It is easy to verify that ϕ is a homomorphism. Additionally, if $a \in \ker \theta$, then $\phi(a + IJ) = \theta(a) = (I, J)$, which shows that $\ker \phi = \{0 + IJ\}$, and thus ϕ is injective. Finally, fix $(y + I, z + J) \in R/I \times R/J$, and again take a and b as elements in I and J , respectively, such that $a + b = 1$. Defining the element $x \in R$ as $x = az + by$, we can see that

$$x \equiv by \equiv (1 - a)y \equiv y - ay \equiv y \pmod{a},$$

as well as

$$x \equiv az \equiv (1 - b)z \equiv z - bz \equiv z \pmod{b}.$$

This shows that

$$\phi(x + IJ) = \theta(x) = (x + I, x + J) = (y + I, z + J),$$

thus ϕ is surjective, hence an isomorphism, and $R/IJ \cong R/I \times R/J$. ■

Lemma 1.8: Let R be a ring and let $I_1, I_2, \dots, I_n$ be ideals of R such that $I_i + I_j = R$ for all $1 \leq i < j \leq n$. We have that $A/I_1 I_2 \cdots I_n \cong A/I_1 \times A/I_2 \times \cdots \times A/I_n$.

Proof: The previous lemma proves the case for $n = 2$. We now use induction on n and assume the case for $n - 1$ holds. Define the ideal $J = I_2 I_3 \cdots I_n$ in R . Note that for $2 \leq k \leq n$, we have $J \subseteq I_k$ and $I_1 + I_k = R$, thus we can find elements $a_k \in I_1$ and $b_k \in J$ such that $a_k + b_k = 1$, and thus

$$1 = \prod_{k=2}^n (a_k + b_k).$$

Using this we obtain the equality $1 = c + a_2 a_3 \cdots a_n$, where c is the sum of the terms in the product that contain at least one b_k . This implies that $c \in I_1$, and thus $1 = c + a_2 a_3 \cdots a_n \in I_1 + J$. Because $I_1 + J$ is an ideal that contains 1, we know that $I_1 + J = R$.

By the previous lemma, we have

$$A/I_1 J \cong A/I_1 \times A/J,$$

and invoking the induction hypothesis, we can see that

$$A/J = A/I_2 I_3 \cdots I_n \cong A/I_2 \times A/I_3 \times \cdots \times A/I_n,$$

thus we have that

$$A/I_1 I_2 \cdots I_n = A/I_1 J \cong A/I_1 \times A/J \cong A/I_1 \times A/I_2 \times \cdots \times A/I_n,$$

completing the proof. ■

Proposition 1.9: Let m and n be relatively prime integers, then we have that $\mathbb{Z}/(mn)\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Proof: By Bezout's lemma, there exist integers x and y such that

$$xm + yn = 1,$$

which means that the ideal $m\mathbb{Z} + n\mathbb{Z}$ contains 1, and thus is equal to $\mathbb{Z}$. Applying **Lemma 1.7**, we obtain

$$\mathbb{Z}/(mn)\mathbb{Z} = \mathbb{Z}/(m\mathbb{Z}n\mathbb{Z}) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

Corollary 1.10: If m and n are relatively prime, then $\phi(mn) = \phi(m)\phi(n)$. ■

Proof: We previously established that $\phi(mn)$ is equal to the number of units in $\mathbb{Z}/mn\mathbb{Z}$, and thus the number of units in $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$. Since an element $(a, b) \in \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ is a unit if and only if a and b are units in their respective rings, we have exactly $\phi(m)\phi(n)$ choices for units in $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, thus proving the equality. ■

Corollary 1.11: For fixed $n \in \mathbb{N}$, let $p_1^{a_1} p_2^{a_2} \cdots p_m^{a_m}$ be its prime factorization, then we have that

$$\phi(n) = n \prod_{k=1}^m \left(1 - \frac{1}{p_k}\right).$$

Proof: Since powers of distinct primes are always relatively prime, we have that $p_i^{a_i}\mathbb{Z} + p_j^{a_j}\mathbb{Z} = \mathbb{Z}$ for all $1 \leq i < j \leq m$. Additionally, $\mathbb{Z}/n\mathbb{Z}$ has $\phi(n)$ units. Furthermore, we have that

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p_1^{a_1}\mathbb{Z} p_2^{a_2}\mathbb{Z} \cdots p_m^{a_m}\mathbb{Z} \cong \mathbb{Z}/p_1^{a_1}\mathbb{Z} \times \mathbb{Z}/p_2^{a_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_m^{a_m}\mathbb{Z},$$

but the number of units in the right hand side is equal to $\phi(p_1^{a_1})\phi(p_2^{a_2}) \cdots \phi(p_m^{a_m})$.

Thus, we have that

$$\begin{aligned}\phi(n) &= \prod_{k=1}^m \phi(p_k^{a_k}) = \prod_{k=1}^m p_k^{a_k} - p_k^{a_k-1} = \prod_{k=1}^m p_k^{a_k} \left(1 - \frac{1}{p_k}\right) = \prod_{k=1}^m p_k^{a_k} \prod_{k=1}^n \left(1 - \frac{1}{p_k}\right) \\ &= n \prod_{k=1}^m \left(1 - \frac{1}{p_k}\right).\end{aligned}$$

■

1.4 Preliminaries on Modules

Proposition 1.12: Let R be a ring, I an indexing set, and M an R -module. Additionally, let $\{m_i\}_{i \in I}$ be a fixed collection of elements in M , and let $a = \{a_i\}_{i \in I}$ be an element in $R^{(I)}$. We have that the function $\phi : R^{(I)} \rightarrow M$, defined as

$$\phi(a) = \sum_{i \in I} a_i m_i,$$

is an R -module homomorphism. Furthermore, the following are true:

- (a) $\{m_i\}_{i \in I}$ is linearly independent if and only if ϕ is injective
- (b) If I is finite, then $\{m_i\}_{i \in I}$ generates M if and only if ϕ is surjective

Proof: Let $a = \{a_i\}_{i \in I}$ and $b = \{b_i\}_{i \in I}$ be elements in $R^{(I)}$. We can see that

$$\phi(a + b) = \sum_{i \in I} (a_i + b_i) m_i = \sum_{i \in I} a_i m_i + \sum_{i \in I} b_i m_i = \phi(a) + \phi(b),$$

and for $r \in R$, we have that

$$\phi(ra) = \sum_{i \in I} r a_i m_i = r \sum_{i \in I} a_i m_i = r \phi(a).$$

Note that we can factor the r out of the sum since only finitely many terms are nonzero. Thus, we have shown that ϕ is an R -module homomorphism.

Next, assume that the m_i are linearly independent. Letting $a, b \in R^{(I)}$ where $\phi(a) = \phi(b)$, we see that

$$\sum_{i \in I} a_i m_i = \sum_{i \in I} b_i m_i,$$

and thus

$$\sum_{i \in I} (a_i - b_i) m_i = 0.$$

The linear independence of the m_i force $a_i - b_i = 0$ for all $i \in I$, hence $a_i = b_i$ and $a = b$, which shows the injectivity of ϕ . Conversely, assume ϕ is injective. It is easy to see that $\phi(0) = \{0\}_{i \in I}$, as

$$\phi(0) = \sum_{i \in I} 0 m_i = 0.$$

Additionally, if we have $a \in R^{(I)}$ such that $\phi(a) = 0$, then by the injectivity of ϕ we have that $a = 0$, and thus $a_i = 0$ for all $i \in I$.

Moving on to (b), we assume that $\{m_i\}_{i \in I}$ generates M . Further, we assume that I is finite. This means that, for all $m \in M$, we can represent it as $m = \sum_{i \in I} a_i m_i$, where the a_i are elements in R . Defining $a = \{a_i\}_{i \in I}$, we easily see that $\phi(a) = m$, showing the surjectivity of ϕ . Finally, assuming ϕ is surjective, we have for any $m \in M$ an element $a \in R^{(I)}$ such that $\phi(a) = m$. This means that

$$m = \sum_{i \in I} a_i m_i,$$

thus each element in M can be represented as a linear combination of the m_i , showing that $\{m_i\}_{i \in I}$ generates M . ■

Lemma 1.13 If T is a partially ordered set, then the following are equivalent:

- (a) Every nonempty subset of T contains a maximal element
- (b) Every nondecreasing sequence of elements of T is stationary

Proof: Fix an increasing sequence of elements in T , say $\{t_n\}_{n \in \mathbb{N}}$, then assuming (a), we can choose t_q as the maximal element of this sequence. Thus, if $n \geq q$, then $t_n \geq t_q$, but the maximality of t_q forces $t_n = t_q$, showing that the sequence is stationary and that (a) $\implies$ (b).

Conversely, assume (b), and let S be any nonempty subset of T . To establish a contradiction, suppose S has no maximal element, then for all $x \in S$, we have some $y \in S$ where $x < y$. Thus, by the axiom of choice, we can define a function $f : S \rightarrow S$ where all x map to strictly larger values. Choosing an arbitrary value $t_0 \in S$, we can define the sequence $\{t_n\}_{n \in \mathbb{N}}$ where $t_n = f(t_{n-1})$. By the construction of f this is a strictly increasing sequence and thus cannot be stationary, a contradiction to our assumption. Thus, we must have that S has a maximal element, proving (b) $\implies$ (a). ■

Theorem 1.14

Fix a ring R and an R -module M , then the following are equivalent:

- (a) Every collection of submodules of M has a maximal element under the ordering of set containment
- (b) Every nondecreasing sequence of submodules of M is stationary
- (c) Every submodule of M has finite type

Proof: We show that (a) $\implies$ (c) $\implies$ (b) $\implies$ (a).

First, assuming (a), we fix a submodule E of M . Denoting Φ as the collection of submodules of E with finite type, we let F be the maximal element of Φ . We can do this since $\{0\}$ is a submodule E , and thus Φ is nonempty. Also note that, for all $x \in E$, we have by the closure of addition and scalar multiplication that $F + Rx$ is a submodule of E , and thus is of finite type, generated by x and any finite set of generators for F . From this it is clear that $F \subseteq F + Ax$, and by our assumption of the maximality of F , we have that $F = F + Ax$, thus $x \in F$. This shows that $E \subseteq F$, and thus $E = F$, showing that E is of finite type, and proving (a) $\implies$ (c).

Now assume (c), and let $\{M_n\}_{n \in \mathbb{N}}$ be a nondecreasing sequence of submodules of M . We define the submodule E of M as follows:

$$E = \bigcup_{n \in \mathbb{N}} M_n.$$

By (c), E is of finite type, and so we can choose a finite set of generating elements for E , say $x_1, x_2, \dots, x_q$. Also, by the definition of E , we have for each $1 \leq n \leq q$ that there exists an index k such that $x_n \in M_k$. Choosing x_n so that k is maximal, we see that $x_n \in M_k$ for all n since the sequence is nondecreasing. Since M_k contains a generating set of E , we have that $M_k = E$. Additionally, if $n \geq k$, then $M_k \subseteq M_n \subseteq E$, showing that $M_k = M_n$ and that the sequence is stationary, proving (c) $\implies$ (b).

Finally, we obtain (b) $\implies$ (a) by noting that it is a special case of **Lemma 1.13**. ■

Corollary 1.15: If R is a principal ideal domain, then every collection of ideals of R contains a maximal element.

Proof: The ideals of R are precisely the submodules of R considered as an R -module over itself. Additionally, as R is a P.I.D., we have that any ideal (submodule) of R is generated by a single element and thus is of finite type. Invoking (c) $\implies$ (a) from **Theorem 1.14**, we have shown that any collection of ideals in R contains a maximal element. ■

1.5 Modules Over Principal Ideal Domains

Theorem 1.16 Let R be a principal ideal domain, M a free R -module of rank n , and N a submodule of M . Then the following are true:

- (a) N is free and has rank $q \leq n$
- (b) If N is not the trivial module, then there exists a basis $b_1, b_2, \dots, b_n$ of M and elements $a_1, a_2, \dots, a_q \in R$ such that $a_1b_1, a_2b_2, \dots, a_qb_q$ is a basis of N , and where a_i divides a_{i+1} for all $1 \leq i \leq q-1$.

Proof: This theorem is apparent for the trivial module, and so we can assume that N is nontrivial. Fix $f \in L(M, R)$ and consider the image $f(N)$. Since R can be interpreted as an R -module over itself, we have that f is an R -module homomorphism, and thus $f(N)$ is a submodule of R . Consequently, $f(N)$ is an ideal in R , but since R is a P.I.D., we have that $f(N) = Ra_f$ for some element $a_f \in R$ that depends on f . By **Corollary 1.15**, we can choose f such that $f(N)$ is maximal among all ideals $g(N)$, $g \in L(M, R)$.

We now choose a basis $x_1, x_2, \dots, x_n$ of M , and define $\pi_i : M \rightarrow R$ as the projection function on the i^{th} coordinate. This function is well defined since the existence of a basis implies the existence of a bijective module homomorphism from M to R^n . Additionally, because N is nontrivial, it has at least one element with a nonzero coordinate, and thus the image $\pi_i(N)$ is a nontrivial submodule of R for at least one i . Note also that $\pi_i \in L(M, R)$, thus our assumption of the maximality of $f(N)$ gives us $\pi_i(N) \subseteq f(N)$, showing that $f(N)$ is a nontrivial submodule (ideal) of R . Consequently, the element a_f is nonzero.

Fix an element $m \in N$ where $f(m) = a_f$. This is possible because $f(N) = Ra_f$ by construction. Let g be an arbitrary linear form in $L(M, R)$ and define $d = \gcd(a_f, g(m))$. Since $d \mid a_f$, we have $Ra_f \subseteq Rd$, and since Ra_f is maximal, we have $Rd \subseteq Ra_f$, proving that $a_f \mid d$, and thus $a_f \mid g(m)$.

A consequence of this is that $a_f \mid \pi_i(m)$, thus we can choose $b_i \in R$ where $\pi_i(m) = a_fb_i$ for all i . Defining $m' = \sum_{i=1}^n b_ix_i$, we can see that

$$a_fm' = \sum_{i=1}^n a_fb_ix_i = \sum_{i=1}^n \pi_i(m)x_i = m.$$

It is also easy to see that $a_ff(m') = f(a_fm') = f(m) = a_f$, thus $a_f[f(m') - 1] = 0$, which implies that $f(m') - 1 = 0$ since a_f is nonzero. We thus have that $f(m') = 1$.

We will now prove the following two assertions:

- (1) $M = \ker f + Rm'$

$$(2) \quad N = (N \cap \ker f) + Rm$$

- (1): For all $x \in M$, we have $x = (x - f(x)m') + f(x)m'$, and it is easy to see that $f(x)m' \in Rm'$. Additionally, we have that $f(x - f(x)m') = f(x) - f(f(x)m') = f(x) - f(x)f(m') = 0$, since $f(m') = 1$. Thus $x - f(x)m' \in \ker f$, and as a result $x \in \ker f + Rm'$. This shows that $M \subseteq \ker f + Rm'$. The containment $\ker f + Rm' \subseteq M$ is true since $\ker f$ and Rm' consist of elements of M , thus proving $M = \ker f + Rm'$.
- (2): Since $f(N) = Ra_f$, we have for all $x \in N$ that $f(x) = ba_f$ for some $b \in R$. This means that $x = (x - f(x)m') + f(x)m' = (x - f(x)m') + ba_fm'$, and thus $x = (x - f(x)m') + bm$. We already showed that $x - f(x)m' \in \ker f$, and combined with the fact that $x - f(x)m' = x - ba_fm' = x - bm \in N$, we have $x - f(x)m' \in N \cap \ker f$. It has thus been shown that $x \in (N \cap \ker f) + Rm$, and $M \subseteq (N \cap \ker f) + Rm$. The converse is clear by recalling the reasoning from (1), thus we have shown (2).

We can now prove (a) by induction on the rank q of N . If $q = 0$, then $N = \{0\}$ and we are finished. Otherwise, if $q > 0$, we have that the rank of $N \cap \ker f$ is $q - 1$ by (2), thus $N \cap \ker f$ is a free R -module by the induction hypothesis. Adding m to a basis of $N \cap \ker f$ we obtain a basis for N , thus showing that N is a free module and proving (a).

Finally, we show (b) by induction on the rank n of M . Again, if $n = 0$, then $M = \{0\}$ and we are finished. For $n > 0$, we similarly find that the rank of $\ker f$ is $n - 1$ using (1). Using the induction hypothesis on $\ker f$ and its submodule $N \cap \ker f$, we obtain a basis $y_2, y_3, \dots, y_n$ of $\ker f$, an integer $q \leq n$, and elements $a_2, a_3, \dots, a_q$, such that $a_2y_2, a_3y_3, \dots, a_qy_q$ is a basis of $N \cap \ker f$, and where $a_i \mid a_{i+1}$ for $2 \leq i \leq q - 1$. We define $a_1 = a_f$ and $y_1 = m'$. By (1), we have that $y_1, y_2, \dots, y_n$ is a basis for M , and by (2) combined with the fact that $m = a_fm'$, we have that $a_1y_1, a_2y_2, \dots, a_qy_q$ is a basis for $N \cap \ker f$. Define the linear form $g : M \rightarrow R$ where $g(y_1) = g(y_2) = 1$ and $g(y_i) = 0$ if $i \geq 3$. We have that

$$a_1 = a_f = a_fg(y_1) = g(a_fm') = g(m) \in g(N),$$

and thus $Ra_f \subseteq g(N)$. The maximality of Ra_f forces $Ra_1 = Ra_f = g(N)$. Since $a_2y_2 \in N \cap \ker f$, we have that $a_2 = g(a_2y_2) \in g(N) = Ra_1$, which means that $Ra_2 \subseteq Ra_1$ and $a_1 \mid a_2$, thus proving (b). $\blacksquare$

Corollary 1.17: Fix a principal ideal ring R , and let M be an R -module of finite type, then there exists a finite chain of ideals $I_1 \supseteq I_2 \supseteq \dots \supseteq I_n$ in R where

$$M \cong R/I_1 \times R/I_2 \times \dots \times R/I_n.$$

Proof: Since M is of finite type, we can choose a finite generating set which induces a surjective homomorphism $\phi : R^n \rightarrow M$. Note that by the first isomorphism theorem for modules, we have that $M \cong R^n / \ker \phi$. The previous theorem allows us to choose a basis $x_1, x_2, \dots, x_n$ of R^n , an integer $q \leq n$, and a basis $a_1x_1, a_2x_2, \dots, a_qx_q$ of $\ker \phi$ where $a_i \mid a_{i+1}$ for $0 \leq i \leq q-1$. We thus have that $R^n \cong Rx_1 \times Rx_2 \times \dots \times Rx_n$. Defining $a_p = 0$ for $q+1 \leq p \leq n$, we obtain $\ker \phi \cong Ra_1x_1 \times Ra_2x_2 \times \dots \times Ra_nx_n$, and thus

$$\begin{aligned} M \cong R^n / \ker \phi &\cong (Rx_1 \times Rx_2 \times \dots \times Rx_n) / (Ra_1x_1 \times Ra_2x_2 \times \dots \times Ra_nx_n) \\ &\cong R/Ra_1 \times R/Ra_2 \times \dots \times R/Ra_n. \end{aligned}$$

Finally, since $a_i \mid a_{i+1}$, we have that $Ra_{i+1} \subseteq Ra_i$, thus completing the proof. $\blacksquare$

Note that the fundamental theorem of finitely generated abelian groups, which states that for any finitely generated abelian group G , there are integers $k_1, k_2, \dots, k_r$ and n where $k_i \mid k_{i+1}$ and

$$G \cong \mathbb{Z}^n \times \mathbb{Z}/\mathbb{Z}k_1 \times \mathbb{Z}/\mathbb{Z}k_2 \times \dots \times \mathbb{Z}/\mathbb{Z}k_r,$$

is a special case of this corollary since any abelian group has a $\mathbb{Z}$ -module structure.

Corollary 1.18: Every torsion free module M of finite type over a principal ideal domain R is free.

Proof: Using the previous result we obtain a chain of ideals $I_1 \supseteq I_2 \supseteq \dots \supseteq I_n$ where

$$M \cong R/I_1 \times R/I_2 \times \dots \times R/I_n.$$

Assume for the sake of establishing a contradiction that I_1 is a nontrivial ideal, then we can choose a nonzero element $a \in I_1$. Fixing a nonzero element $x + I_1 \in R/I_1$, we note that $(x + I_1, 0, \dots, 0)$ is an element in M . Finally, since ax is an element of I_1 , we have that $ax + I_1 = I_1$, and thus $a(x + I_1, 0, \dots, 0) = (I_1, 0, \dots, 0) = 0$, violating the assumption that M is torsion free. We thus have that I_1 is the trivial ideal, which implies that I_i is trivial for all i since each ideal is contained in I_1 . This proves that $M \cong R^n$, thus permitting the basis $(1, 1, \dots, 1)$ and proving that M is free. $\blacksquare$

Corollary 1.19: Let R be a principal ideal domain, then any R -module M of finite type is isomorphic to a finite product of R -modules $M_1 \times M_2 \times \dots \times M_n$, where M_i is either equal to R or equal to the quotient R/Rp^k , where p^k is a power of some prime element in R .

Proof: Using **Corollary 1.17**, we find a chain of ideals $I_1 \supseteq I_2 \supseteq \dots \supseteq I_n$ of R where M is isomorphic to the product of the quotients of R by these ideals. Now, consider R/I_i for a fixed i . If I_i is trivial, $R/I_i = R$ and we are finished, so we can assume that I_i is nontrivial. We know that $I_i = Ra$ for some $a \in R$, and since R is a unique

factorization domain, we can choose unique prime elements $p_1, p_2, \dots, p_r \in R$, positive integers $k_1, k_2, \dots, k_r$, and a unit $u \in R$ where $a = up_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$. Additionally, since $\gcd(p_i^{k_i}, p_j^{k_j}) = 1$ if $i \neq j$, we have that $Rp_i^{k_i} + Rp_j^{k_j} = R$. Finally, we note the fact that

$$Ra = Rup_1^{k_1} p_2^{k_2} \cdots p_r^{k_r} = Rp_1^{k_1} Rp_2^{k_2} \cdots Rp_r^{k_r},$$

thus we can invoke **Lemma 1.8** to see the following:

$$R/Ra \cong R/Rp_1^{k_1} \times R/Rp_2^{k_2} \times \cdots \times R/Rp_r^{k_r}.$$

This shows that the product of the R/I_i consists of factors of R or R/Rp^k . ■

Corollary 1.20: Let G be a finite abelian group. There exists an element $x \in G$ whose order is the least common multiple of the orders of the other elements in G .

Proof: Since abelian groups have an induced $\mathbb{Z}$ -module structure, we invoke **Corollary 1.17** to obtain a chain of ideals $\mathbb{Z}a_1 \supseteq \mathbb{Z}a_2 \supseteq \cdots \supseteq \mathbb{Z}a_n$ where

$$G \cong \mathbb{Z}/\mathbb{Z}a_1 \times \mathbb{Z}/\mathbb{Z}a_2 \times \cdots \times \mathbb{Z}/\mathbb{Z}a_n.$$

We thus have that $a_i \mid a_{i+1}$ for $1 \leq i \leq n-1$. Additionally, we know that $a_i \neq 0$ for all i , otherwise $\mathbb{Z}$ would be a factor of G , violating the assumption that G is finite. Letting $x \in \mathbb{Z}/\mathbb{Z}a_n$ where $x \equiv 1 \pmod{a_n}$, we note that $(0, 0, \dots, 0, x)$ is an element in G and has order a_n . Fixing an arbitrary element $y = (y_1, y_2, \dots, y_n) \in G$, we can see that $a_n y_i = 0$ for all i since $a_i \mid a_n$. We thus have that $a_n y = 0$, which shows that a_n is a multiple of the orders of all elements in G . Finally, since a_n is itself the order of an element, we know it is the least common multiple of these orders. ■

1.6 Roots of Unity in a Field

Theorem 1.21: Let F be a field. If G is a finite subgroup of F^* , then G consists of roots of unity and is cyclic.

Proof: By **Corollary 1.20**, there exists an element $x \in G$ whose order, say n , is a multiple of the order of all elements in G . This means that every element in G satisfies the polynomial equation $X^n = 1$, and thus every element is a root of unity. Since the polynomial $X^n - 1$ has at most n roots, we know that the order of G is at most n . We can also see that G consists of the n unique elements $x, x^2, \dots, x^{n-1}, x^n = 1$, thus G is generated by x and is cyclic. ■

1.7 Finite Fields

Proposition 1.22: If F is a field with characteristic $p \neq 0$, then $px = 0$ for all $x \in F$, and $(x + y)^p = x^p + y^p$ for all $x, y \in F$.

Proof: The binomial theorem gives us

$$(x + y)^p = \sum_{k=0}^p \binom{p}{k} x^k y^{p-k} = x^p + y^p + \sum_{k=1}^{p-1} \binom{p}{k} x^k y^{p-k}.$$

Since $\binom{p}{k} = p!/k!(p-k)!$, and since p does not divide $k!(p-k)!$, we have that it is a multiple of p , and thus each term in the $\sum_{k=1}^{p-1} \binom{p}{k} x^k y^{p-k}$ is a multiple of p , and thus equal to 0 since F has characteristic p . This shows that $(x + y)^p = x^p + y^p$, completing the proof. $\blacksquare$

Theorem 1.23: Let F be a finite field with cardinality q , then the following are true:

- (a) The characteristic p of F is prime, F is a vector space of finite dimension k , and $q = p^k$
- (b) The group F^* is cyclic and has order $q - 1$
- (c) $x^{q-1} = 1$ for all $x \in F^*$, and $x^q = x$ for all $x \in F$.

Proof: We know that F cannot contain $\mathbb{Z}$, since F is finite, thus it must contain $\mathbb{Z}/p\mathbb{Z}$ for some prime p . F also inherits a $\mathbb{Z}/p\mathbb{Z}$ -module structure from the unique ring homomorphism $\phi : \mathbb{Z} \rightarrow F$, and since $\mathbb{Z}/p\mathbb{Z}$ is a field, F is a vector space. Additionally, since F is finite, it must have finite dimension k , which means that $F \cong (\mathbb{Z}/p\mathbb{Z})^k$. Since $(\mathbb{Z}/p\mathbb{Z})^k$ has p^k elements, we have shown that $q = p^k$, thus proving (a). (b) follows from **Theorem 1.21** since F^* is finite, and (c) is immediate from (b). $\blacksquare$

Theorem 1.24: Fix a finite field F and let f be a homogeneous polynomial of degree d in $F[X_1, X_2, \dots, X_n]$. If $d < n$, then there exists a point $(x_1, x_2, \dots, x_n) \in K^n$ that is not the origin where $f(x_1, x_2, \dots, x_n) = 0$. In other words, every homogeneous polynomial with coefficients in F has a nontrivial root in F .

Proof: Denote q as the cardinality of F and p the characteristic, then q is a power of p . Let V be the set of zeroes of f in K^n , i.e. the set

$$\{(x_1, x_2, \dots, x_n) \in K^n : f(x_1, x_2, \dots, x_n) = 0\}.$$

For all $x \in K^n \setminus V$, we have by the previous theorem that $f(x)^{q-1} = 1$, otherwise x is a root and $f(x)^{q-1} = 0$. Denoting $g(x) = f(x)^{q-1}$, we have shown that g is the

characteristic function $\chi_{K^n \setminus V}$. We can also see that the number of points in $K^n \setminus V$, modulo p , is given by the sum

$$\sum_{x \in F^n} g(x). \quad (1)$$

If this sum is 0, then $|F^n \setminus V|$ is a multiple of p . Also note that $|F^n|$ is a multiple of p , and thus $|V|$ is a multiple of p since $|V| = |F^n| - |F^n \setminus V|$. Because p divides $|V|$, and because the origin is clearly a root of f , we have that $|V| \geq p \geq 2$, and thus V must have a nontrivial root for f .

Consequently, it now suffices to show that the sum in (1) is 0. Since g is a linear combination of monomials $m(X_1, X_2, \dots, X_n) = X_1^{a_1} X_2^{a_2} \cdots X_n^{a_n}$, it also suffices to show that

$$\sum_{x \in F^n} m(x) = \sum_{x \in F^n} x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n} = \sum_{x_1 \in F} x_1^{a_1} \sum_{x_2 \in F} x_2^{a_2} \cdots \sum_{x_n \in F} x_n^{a_n} = 0$$

for any monomial $m \in F[X_1, X_2, \dots, X_n]$, which further reduces to the problem of solving $\sum_{x \in F} x^k$ where $k \in \mathbb{N}_0$. If $k = 0$, then $x^k = x^0 = 1$ for all $x \in F$, and thus

$$\sum_{x \in F} x^k = \sum_{x \in F} 1 = q \equiv 0 \pmod{p},$$

so we can take $k > 0$, in which case we have that $0^k = 0$, thus we may restrict the sum to F^* . F is finite, and so F^* is a cyclic group of order $q - 1$. Choosing a generator $y \in F^*$, we have that the powers of y from 0 to $q - 2$ cover the elements in F^* , and so

$$\sum_{x \in F^*} x^k = \sum_{i=0}^{q-2} (y^i)^k = \sum_{i=0}^{q-2} y^{ki},$$

which is a geometric progression. We now have two cases:

(a) If $y^k \neq 1$, then k is not a multiple of $|y| = q - 1$, and thus

$$\sum_{i=0}^{q-2} y^{ki} = \frac{1 - y^{k(q-1)}}{1 - y^k} = \frac{1 - 1}{1 - y^k} = 0.$$

(b) If $y^k = 1$, then k is a multiple of $q - 1$, and so

$$\sum_{i=0}^{q-2} y^{ki} = \sum_{i=0}^{q-2} 1 = q - 1.$$

Thus, the sum (1) is 0 unless the a_i in the representation of m are all multiples of $q - 1$. In this case, the degree of m would be at least $(q - 1)n$. However, $\deg g(x) =$

$\deg f(x)^{q-1} = (q-1)d < (q-1)n$, and so g fits the case of (a), and thus (1) is 0, completing the proof. ■